

Md Mobarak Karim

Optical Systems Engineer | Biomedical Imaging Scientist

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Professional Summary

Optical Systems Engineer and Biomedical Imaging Scientist with a Ph.D. in Biomedical Engineering and 6+ years of experience in optical design, microscopy, system integration, automation, and quantitative imaging. Current Postdoctoral Associate in **Dr. Rebecca Richards-Kortum's** laboratory at Rice University, developing optical and computational imaging technologies for AccessPath and PrecisionView.

Technical Skills

Optical Systems	OCT/OCM, one-photon and two-photon light-sheet microscopy, MUSE microscopy, fluorescence microscopy, multiphoton/confocal microscopy, laser-scanning microscopy
Optical Engineering	Zemax OpticStudio, lens/relay design, 4f systems, beam delivery, galvos, polarization, fiber optics, phase-mask optics, optical alignment
Characterization	Resolution, PSF, MTF, SNR/CNR, sensitivity, roll-off, FOV, DOF, aberrations, calibration, repeatability, design verification
Software / Imaging	Python, MATLAB, LabVIEW, C/C++, GUI development, XYZ automation, autofocus, stitching, background correction, averaging, virtual H&E, segmentation, co-registration

Professional and Research Experience

Postdoctoral Associate - Optical Systems and Computational Microscopy

Aug 2026–Present

Richards-Kortum Laboratory, Department of Bioengineering, Rice University
Principal Investigator: Dr. Rebecca Richards-Kortum

Houston, TX

- **Design AccessPath optics in Zemax** for lens/relay development, component selection, integration, and imaging characterization.
- **Built an automated XYZ unit-microscopy and GUI** integrating synchronized acquisition, autofocus, averaging, background correction, stitching, and parallel virtual H&E processing.
- **Leading AccessPath APHA-H Phase II optics activities**, including array-microscopy architecture and **phase-mask design, fabrication, and validation**.
- Support **PrecisionView optics** through alignment, characterization, troubleshooting, and performance optimization.
- Evaluate **contrast mechanisms and wavelength combinations for virtual H&E** to improve tissue contrast and suppress optical background.

Graduate Research Assistant - Multimodal Optical Systems

Jan 2021–May 2026

Biomedical Optics Laboratory, University of Houston

Houston, TX

- Designed and integrated a co-registered multimodal imaging platform combining a **1051-nm swept-source OCT system** (109-nm bandwidth, 100-kHz sweep rate) with **one-photon and two-photon LSFM** for simultaneous structural and molecular imaging of mouse and zebrafish embryos.
- Achieved approximately **7–7.6 micrometer OCT axial resolution, 15 micrometer lateral resolution**, approximately **100 dB sensitivity**, and approximately **6.9 dB sensitivity roll-off over 4.13 mm**; characterized system resolution, sensitivity, roll-off, field of view, image contrast, and repeatability.
- Engineered beam delivery, polarization control, galvo scanning, scan-lens interfaces, acquisition timing, camera/laser synchronization, and sample positioning for stable multimodal 3D imaging.
- Demonstrated one-photon fluorescence penetration of approximately **250 micrometers** and two-photon visualization beyond **800 micrometers** in scattering embryonic tissue while OCT provided label-free whole-embryo morphology.
- Developed quantitative OCT algorithms for depth-resolved attenuation-coefficient estimation with noise subtraction, confocal correction, and sensitivity-roll-off compensation; validated the method using multilayer TiO₂ phantoms.
- Built MATLAB and Python pipelines for OCT reconstruction, 3D visualization, multimodal co-registration, morphometry, image-quality analysis, automated quality control, and standardized data processing.
- Developed cost-effective (300-500 USD) imaging incubator for the OCT-LS system.
- Mentored **three Ph.D. students and several undergraduate researchers** in OCT/LSFM system alignment, optical troubleshooting, image acquisition, MATLAB/Python analysis, experimental design, and technical presentation.
- Led optical hardware development, software integration, biological imaging, data analysis, manuscript preparation, and coordination across multidisciplinary research teams.

Graduate Research Assistant - Optical Design and Two-Photon Microscopy

Aug 2018–Dec 2020

3D Biomedical Imaging and Technology Laboratory, GIST

Gwangju, South Korea

- Designed and optimized broadband scan and tube lenses for laser-scanning microscopy using **Zemax OpticStudio**. Evaluated **wavefront error, Strehl ratio, aberrations, field of view, telecentricity, and F-theta distortion** across 680–1600 nm.

- Developed four optimized scan lenses and one tube lens, reducing **F-theta distortion from 13.00 to 1.94 and telecentric distortion from 3.52 to 0.4367**.
- Supported development and qualification of a two-photon laser-scanning microscope through optical-component evaluation, alignment, testing, measured-performance analysis, and structured troubleshooting.

Education

Ph.D., Biomedical Engineering

Jan 2021–May 2026

University of Houston

Houston, TX

Advisor: Dr. Kirill V. Larin. Dissertation: *Co-registered optical coherence tomography and light-sheet fluorescence microscopy for structural and molecular imaging of embryonic development*.

M.S., Biomedical Science and Engineering

Aug 2018–Dec 2020

Gwangju Institute of Science and Technology

Gwangju, South Korea

Thesis: *Analysis of an achromatic doublet as a scan lens for laser-scanning microscopy*.

B.S., Electrical and Electronic Engineering

Aug 2013–Aug 2017

International Islamic University Chittagong

Chittagong, Bangladesh

Advanced Training

- Multiscale Biophotonics Training Program, Beckman Laser Institute, **University of California, Irvine**, 2024 - NIH-supported training in fluorescence lifetime imaging, nonlinear microscopy, spectroscopy, and translational biophotonics.
- CLIMB Biophotonics Summer School, Beckman Institute, **University of Illinois Urbana-Champaign**, 2023 - hands-on OCT/OCM system construction, reconstruction, and quantitative imaging.
- **Siegman International School on Lasers, Poland**, 2022 - laser physics, ultrafast optics, nonlinear optics, and advanced photonic systems.

Selected Publications

- **Md Mobarak Karim**, et al. “Multimodal optical coherence tomography and two-photon light-sheet fluorescence microscopy for embryo imaging.” *Journal of Biomedical Optics* (2025). Developed a co-registered structural-molecular platform integrating OCT and two-photon LSFM for intact-embryo imaging.
- **Md Mobarak Karim**, et al. “Depth-resolved attenuation coefficient estimation during embryonic brain development.” *Journal of Biophotonics* (2025). Established quantitative OCT attenuation mapping for region-specific assessment of murine brain development.
- Leah A. Lewis, **Md Mobarak Karim**, et al. “Multimodal Optical Imaging for the Assessment of the Teratogenic Effects of Embryonic Ethanol Exposure on Zebrafish Development.” *Journal of Biomedical Optics* (2026). Integrated OCT and LSFM to quantify morphological and vascular developmental-toxicity phenotypes.
- James L. Klein, **Md Mobarak Karim**, Christian Zevallos-Delgado, Manmohan Singh, Salavat R. Aglyamov, and Kirill V. Larin. “Tartrazine-Based Optical Clearing for Monitoring Embryogenesis with Optical Coherence Tomography.” Submitted to *Optics Letters*. Demonstrated approximately twofold deeper OCT imaging of murine embryos using tartrazine-based optical clearing.
- **Md Mobarak Karim**, et al. “Long-term imaging of zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy.” Manuscript in preparation.

Awards and Honors

2025	Cover image featured in <i>Journal of Biomedical Optics</i>
2024	NIH Travel Grant, UCI Multiscale Biophotonics Training Program
2022	Zuegel Family Scholarship, Siegman International School on Lasers
2022	OPTICA Siegman International School on Lasers Grant
2022	SPIE Photonics West Travel Grant
2020	Outstanding Poster Award, Biomedical Research Meeting, South Korea
2018–2020	Korean Government Scholarship for M.S. study at GIST
2013–2017	Ungraduate Merit Award and Scholarship, Dept. of EEE, IUC

Professional Service and Leadership

- Reviewer for biomedical optics and photonics journals.
- Senior Judge, Texas Science and Engineering Fair - Biomedical Division (2022).
- Executive Member, University of Houston SPIE Student Chapter (2021–Present).
- Member, OPTICA (formerly the Optical Society of America) (2021–Present).
- Founder and former executive member, Bangladesh Robotics Foundation.
- Troop Leader, Bangladesh Red Crescent Youth; Leadership Graduate, BYLC Art of Leadership Program (2016).